

Static Mean-Response Rank Implies Exact Dimension Complexity

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1 Theoretical Setup

1.1 Problem Setting

Let $\mathcal{X}$ be an arbitrary, possibly empty or infinite, domain, and let $\mathcal{H} \subseteq \{-1, +1\}^{\mathcal{X}}$. Write $\mathcal{P}(\mathcal{X})$ for the probability laws on $\mathcal{X}$; in particular, all finitely supported laws and all point masses on a nonempty domain are available. Protocol maps are assumed measurable whenever needed for the expectations below. For $\mathcal{D} \in \mathcal{P}(\mathcal{X})$, $h \in \mathcal{H}$, and $g : \mathcal{X} \rightarrow \{-1, +1\}$, set

$$\mathcal{L}_{\mathcal{D},h}(g) := \Pr_{x \sim \mathcal{D}}[g(x) \neq h(x)].$$

We use exact sign representation only with nonzero scores. Thus $\varphi : \mathcal{X} \rightarrow \mathbb{R}^d$ represents h with weight $w_h \in \mathbb{R}^d$ when $h(x)\langle w_h, \varphi(x) \rangle > 0$ for every x . With the infimum of the empty set understood as $+\infty$, define

$$\text{dc}(\mathcal{H}) := \inf \left\{ d \in \mathbb{N}_0 : \begin{array}{l} \exists \varphi : \mathcal{X} \rightarrow \mathbb{R}^d \quad \forall h \in \mathcal{H} \quad \exists w_h \in \mathbb{R}^d \quad \forall x \in \mathcal{X}, \\ h(x)\langle w_h, \varphi(x) \rangle > 0 \end{array} \right\}.$$

Represent all randomness of one fixed learner A by a seed $U \sim \mu_A$ on a seed space Ω_A . Conditional on $U = u$, at each round $t \leq m$, the learner either stops or selects, as a function of u and the preceding public query-response transcript, an unrestricted bounded query

$$q_t : \mathcal{X} \times \{-1, +1\} \rightarrow [-1, 1].$$

After at most m queries it returns a binary predictor. The seed contains all query-selection and output randomness.

Put $I_\tau = [-1 - \tau, 1 + \tau]$. A deterministic complete response rule R assigns a value in I_τ to every public query-bearing history $(q_1, v_1, \dots, q_{t-1}, v_{t-1}, q_t)$ admitted by A , including histories not reached for a particular seed or preceding reply sequence. It observes the public queries and replies but not the hidden seed except through that public transcript. Let $\mathfrak{R}_A^{\text{all}}$ denote all such rules. For $m = 0$, this family contains the unique empty rule. For $u \in \Omega_A$ and $R \in \mathfrak{R}_A^{\text{all}}$, let $g_{u,R} : \mathcal{X} \rightarrow \{-1, +1\}$ be the terminal predictor.

For $\mathcal{D} \in \mathcal{P}(\mathcal{X})$ and $h \in \mathcal{H}$, call R $(\mathcal{D}, h, \tau)$ -valid if, at every round reached by every seed interacting with R ,

$$|v_t - \mathbb{E}_{x \sim \mathcal{D}} q_t(x, h(x))| \leq \tau.$$

Write $\mathfrak{R}_{A,\tau}(\mathcal{D}, h)$ for these valid complete rules. This is an adaptive adversarial policy class: the accuracy premise below is pointwise in R , and no distribution over responses is introduced. Any randomized response policy is covered conditionally on each realization of its own randomness.

For every complete rule define its pre-instance, seed-averaged terminal response

$$F_R(x) := \mathbb{E}_{U \sim \mu_A}[g_{U,R}(x)] \in [-1, 1]$$

and the static mean-response space and rank

$$V_A := \text{span}_{\mathbb{R}}\{F_R : R \in \mathfrak{R}_A^{\text{all}}\} \subseteq \mathbb{R}^{\mathcal{X}}, \quad r_A := \dim V_A \in \mathbb{N}_0 \cup \{\infty\}.$$

These objects are fixed before any distribution, target, valid policy, or realized seed is selected. When $r_A < \infty$, define in the unique finite-dimensional topology of V_A

$$K_A := \overline{\text{conv}}\{F_R : R \in \mathfrak{R}_A^{\text{all}}\} \subseteq V_A.$$

1.2 Assumptions

Assumption 1 (Primitive parameter regime). *The parameters satisfy $m \in \mathbb{N}_0$, $\tau > 0$, and $\varepsilon \in [0, 1/4)$. The numerical constants $B, k \in \mathbb{R}$ satisfy $B \geq 1$ and $k \geq 1$, are fixed before the learner instance, and are fixed independently of $\mathcal{X}, \mathcal{H}, m, \tau, \varepsilon, \mathcal{D}, h$, all response rules, and all learner seeds.*

Assumption 2 (Fixed randomized adaptive unrestricted-SQ interface). *The learner A , fixed before $\mathcal{D}, h$, and the response policy, has seed law μ_A , makes at most m adaptive bounded unrestricted queries of the form above, may use the complete preceding real-valued public transcript, and returns a binary predictor. Interaction with every deterministic complete response rule is well defined, and all seed averages and displayed population expectations exist. No additional restriction is imposed on the query family, adaptivity, response precision, seed space, or terminal-predictor span.*

Assumption 3 (Every-valid-policy universal guarantee). *For every $\mathcal{D} \in \mathcal{P}(\mathcal{X})$, every $h \in \mathcal{H}$, and every $R \in \mathfrak{R}_{A,\tau}(\mathcal{D}, h)$,*

$$\mathbb{E}_{U \sim \mu_A} \mathcal{L}_{\mathcal{D},h}(g_{U,R}) \leq \varepsilon.$$

The response rule is universally quantified and adversarial; the sole averaging is over U .

Assumption 4 (Static polynomial mean-response-rank certificate). *The rank over all deterministic complete response rules is finite and obeys*

$$r_A = \dim \text{span}_{\mathbb{R}}\{F_R : R \in \mathfrak{R}_A^{\text{all}}\} \leq B(1 + m/\tau^2)^k.$$

This certificate is fixed before $\mathcal{D}, h$, a selected valid rule, and the learner seed. It constrains only the seed-averaged responses F_R , not the cardinality or span of the individual terminal predictors, and it does not assume a representation of $\mathcal{H}$.

1.3 Formalized Goal

The objective is to choose one basis of V_A , independently of every learning instance, whose coordinate map represents every $h \in \mathcal{H}$ with deterministic pointwise signed margin $1 - 2\varepsilon$, and to prove the exact dimension chain

$$\text{dc}(\mathcal{H}) \leq r_A \leq B(1 + m/\tau^2)^k.$$

Only the target weight may depend on h . The desired statement is conditional on Assumption 4, has no probability qualification or domain-size dependence, and must cover empty domains or classes, zero rank, $m = 0$, $\varepsilon = 0$, $B = 1$, and $k = 1$.

2 Preliminaries

Set

$$\rho := 1 - 2\varepsilon.$$

This setting-derived quantity is the signed correlation and pointwise margin appearing in the main theorem. Assumption 1 gives $\rho \in (1/2, 1]$. We use the standard conventions that $\mathbb{R}^0$ contains the unique empty vector, an empty basis spans the zero vector space, and an intersection over an empty family of subsets of a fixed ambient set equals that ambient set. These conventions are needed only to state our result uniformly across the zero-rank and empty-domain cases.

For a finite-dimensional V_A , a basis $\psi_1, \dots, \psi_{r_A}$ determines the coordinate map

$$\varphi_A(x) := (\psi_1(x), \dots, \psi_{r_A}(x)).$$

The proof constructs this basis before any instance is selected. The map is the sole feature map used in the main theorem; proof-only evaluation coordinates used to establish compactness are introduced locally in the appendix.

3 Main Theorem

Theorem 3.1 (Conditional static mean-response-rank theorem). *Under Assumptions 1–4, there is a basis $\psi_1, \dots, \psi_{r_A}$ of V_A , chosen once from the fixed domain, learner, seed law, horizon, tolerance, complete-response interface, and a fixed basis choice, such that the deterministic map*

$$\varphi_A(x) := (\psi_1(x), \dots, \psi_{r_A}(x)) \in \mathbb{R}^{r_A}$$

is independent of every $\mathcal{D}$, target, tolerance-valid response policy, and realized learner seed. For every $h \in \mathcal{H}$, there is a weight $w_h \in \mathbb{R}^{r_A}$ satisfying

$$h(x)\langle w_h, \varphi_A(x) \rangle \geq 1 - 2\varepsilon = \rho > \frac{1}{2} > 0 \quad \text{for every } x \in \mathcal{X}.$$

Consequently,

$$\text{dc}(\mathcal{H}) \leq r_A \leq B(1 + m/\tau^2)^k.$$

The conclusion is deterministic, pointwise, and fixed-horizon, with no hidden constants or domain-cardinality dependence. If $\mathcal{X} = \emptyset$ or $\mathcal{H} = \emptyset$, then $\text{dc}(\mathcal{H}) = 0$. If both are nonempty, then $r_A \geq 1$. At $m = 0$, the bound specializes to $r_A \leq B$; at $\varepsilon = 0$, the signed margin is exactly 1; and $B = k = 1$ gives $r_A \leq 1 + m/\tau^2$.

The theorem is conditional: Assumption 4 is an additional static protocol-family certificate. The argument does not derive that certificate from m and τ alone and does not establish the unconditional universal linear dimension bound. Those source-level questions remain open.

4 Proof Sketch

First form the seed-averaged terminal responses over all complete public response rules, and choose a basis of their finite-dimensional span before selecting any learning instance. Lemma A.1 and Propositions A.1 and A.2 make this construction precise and separate mean-response rank from the generally much larger span of individual seed-specific predictors.

Next choose finitely many point evaluations that coordinatize V_A . They put every raw mean response in a common bounded set, so the closed convex body K_A is compact in one fixed topology

and every point evaluation is continuous. For each distribution-target pair, the exact-center complete rule is tolerance-valid on every reached seed path. The universal guarantee and the binary loss identity then produce a member of K_A with signed correlation at least ρ .

For a nonempty finite set S , play the bilinear payoff $L_h(f, p) = \sum_{x \in S} p(x) h(x) f(x)$ on the fixed body K_A and the simplex $\Delta(S)$. Sion's theorem (Sion, 1958, Theorem 3.4) changes the order of optimization, while the exact-center witness may vary with p inside the same fixed body. This yields one $f_{h,S}$ with margin at least ρ at every point of S . The resulting pointwise constraint sets are closed subsets of the same compact K_A and have the finite-intersection property, including the empty subfamily. Compactness therefore gives a single f_h satisfying all points of an arbitrary domain.

Finally, exact basis coordinates turn f_h into w_h with zero score residual. Strict positivity makes r_A admissible in the definition of dimension complexity, and the primitive rank certificate gives the displayed polynomial upper bound. The appendix proves each of these statements in full, including the empty, zero-rank, zero-query, noiseless, and parameter-boundary cases.

References

- FOLLAND, G. B. (1999). *Real Analysis: Modern Techniques and Their Applications*. 2nd ed. John Wiley & Sons, New York. [13](#)
- SION, M. (1958). On general minimax theorems. *Pacific Journal of Mathematics* **8** 171–176. Doi:10.2140/pjm.1958.8.171. [4](#), [14](#)

A Proof Details

A.1 Complete Responses and Pre-Instance Coordinates

Lemma A.1 (Complete public response rules and the zero-query convention). *Under Assumptions 1 and 2, let $\mathcal{Q}_A$ be the set of all public query-bearing histories*

$$\eta_t = (q_1, v_1, \dots, q_{t-1}, v_{t-1}, q_t), \quad 1 \leq t \leq m,$$

admitted by the fixed learner for some seed and some preceding public reply sequence in I_τ , including histories not reached for a particular seed or reply sequence. Then

$$\mathfrak{R}_A^{\text{all}} = I_\tau^{\mathcal{Q}_A} = \{R : \mathcal{Q}_A \rightarrow I_\tau\}$$

is exactly the family of deterministic complete public-transcript response rules and is nonempty. Every $R \in \mathfrak{R}_A^{\text{all}}$ and $u \in \Omega_A$ determine a terminal predictor $g_{u,R} : \mathcal{X} \rightarrow \{-1, +1\}$. If $m = 0$, then $\mathcal{Q}_A = \emptyset$ and $\mathfrak{R}_A^{\text{all}} = \{R_\emptyset\}$, where $R_\emptyset$ is the unique empty response rule.

Proof. The set $\mathcal{Q}_A$ is determined solely by the fixed public interaction interface: its elements contain public queries and replies but no hidden-seed coordinate. A total function on $\mathcal{Q}_A$ therefore observes the seed only through information already present in the public transcript. Because its domain contains every admitted query-bearing history, including those not reached on one seed path, such a function is complete. Conversely, every deterministic complete response rule assigns one value in I_τ to every element of $\mathcal{Q}_A$, proving the equality of the two families.

Assumption 1 gives $\tau > 0$, and hence

$$0 \in [-1 - \tau, 1 + \tau] = I_\tau.$$

Thus the assignment $R^{\text{zero}}(\eta) = 0$ for all $\eta \in \mathcal{Q}_A$ proves nonemptiness. This rule need not be valid for a later learning instance; it is only an all-rule witness.

For fixed u and R , the learner selects its next action from u and the preceding public transcript. Whenever it issues a query, totality of R supplies the next reply. Assumption 2 makes this recursion well defined and forces stopping after at most m queries with a binary predictor $g_{u,R}$.

If $m = 0$, there is no index t with $1 \leq t \leq m$, so $\mathcal{Q}_A = \emptyset$. There is exactly one function from the empty set to I_τ , namely $R_\emptyset$. The learner makes no query, but its seed-dependent terminal output remains well defined. $\square$

Proposition A.1 (Finite seed-averaged response space and its exact scope). *Under Assumptions 1, 2, and 4, and Lemma A.1, define*

$$F_R(x) := \int_{\Omega_A} g_{u,R}(x) \mu_A(du), \quad V_A := \text{span}_{\mathbb{R}}\{F_R : R \in \mathfrak{R}_A^{\text{all}}\}, \quad r_A := \dim V_A.$$

Then every F_R is a well-defined member of $[-1, 1]^{\mathcal{X}}$, and

$$r_A \leq B(1 + m/\tau^2)^k < \infty.$$

This average-before-span construction gives no bound by r_A on the span of the seed-specific functions $g_{u,R}$. If $\mathcal{X} = \emptyset$, then $V_A = \{0\}$ and $r_A = 0$. If $m = 0$, then

$$V_A = \text{span}_{\mathbb{R}}\{F_{R_\emptyset}\}, \quad r_A \in \{0, 1\}, \quad r_A \leq B.$$

Proof. Lemma A.1 supplies a terminal binary function $g_{u,R}$ for every seed and complete rule. The measurability clause of Assumption 2 makes the displayed seed integral well defined. Since $-1 \leq g_{u,R}(x) \leq 1$, monotonicity of integration against the probability law μ_A gives

$$-1 \leq \int_{\Omega_A} g_{u,R}(x) \mu_A(du) \leq 1.$$

Thus $F_R : \mathcal{X} \rightarrow [-1, 1]$, and finite real linear combinations of these functions form $V_A \subseteq \mathbb{R}^{\mathcal{X}}$.

Assumption 4 applies to exactly this space:

$$r_A = \dim \text{span}_{\mathbb{R}} \left\{ x \mapsto \int_{\Omega_A} g_{u,R}(x) \mu_A(du) : R \in \mathfrak{R}_A^{\text{all}} \right\} \leq B(1 + m/\tau^2)^k.$$

The right-hand side is finite because B, k are finite, $m \in \mathbb{N}_0$, and $\tau > 0$. Hence $r_A \in \mathbb{N}_0$.

To make the scope of this certificate explicit, define

$$W_A^{\text{seed}} := \text{span}_{\mathbb{R}} \{g_{u,R} : u \in \Omega_A, R \in \mathfrak{R}_A^{\text{all}}\}.$$

The two operations are genuinely different:

$$V_A = \text{span}_R \left\{ \int g_{u,R} d\mu_A(u) \right\}, \quad W_A^{\text{seed}} = \text{span}_{u,R} \{g_{u,R}\}.$$

No individual $g_{u,R}$ is claimed to lie in V_A , and no finiteness or cardinality assertion about W_A^{seed} is used below.

The distinction can be arbitrarily large even under all four assumptions. Fix $N \geq 5$, take $\mathcal{X}_N = \{1, \dots, N\}$, $m = 0$, $B = k = 1$, arbitrary $\tau > 0$, and $\varepsilon = 1/N < 1/4$. Let $\mathcal{H} = \{h_+\}$, where $h_+(i) = +1$, let U be uniform on $\{1, \dots, N\}$, and let the zero-query learner with seed j return

$$g_{j,R_\emptyset}(i) = \begin{cases} -1, & i = j, \\ +1, & i \neq j. \end{cases}$$

The unique mean response is the nonzero constant

$$F_{R_\emptyset}(i) = \frac{N-1}{N} - \frac{1}{N} = 1 - \frac{2}{N},$$

so $r_A = 1$ and the mean-response-rank certificate holds with equality. The N terminal predictors are nevertheless linearly independent. If $\sum_{j=1}^N c_j g_{j,R_\emptyset} = 0$ and $S_c = \sum_{j=1}^N c_j$, evaluation at i gives $S_c - 2c_i = 0$ for every i . Hence $c_i = S_c/2$ and $S_c = NS_c/2$. Since $N \neq 2$, this forces $S_c = 0$ and all $c_i = 0$. Thus $\dim W_A^{\text{seed}} = N$ while $r_A = 1$.

This example also satisfies Assumption 3. For every probability law $\mathcal{D}$ on $\mathcal{X}_N$, the only response rule is $R_\emptyset$, and

$$\mathbb{E}_U \mathcal{L}_{\mathcal{D}, h_+}(g_{U, R_\emptyset}) = \frac{1}{N} \sum_{j=1}^N \mathcal{D}(\{j\}) = \frac{1}{N} = \varepsilon.$$

Thus the gap between the two spans is compatible with the complete theorem contract.

If $\mathcal{X} = \emptyset$, there is only one real-valued function on $\mathcal{X}$, namely the zero vector of $\mathbb{R}^{\mathcal{X}}$. Every $g_{u,R}$ and F_R is this function, so $V_A = \{0\}$ and $r_A = 0$. If $m = 0$, Lemma A.1 gives the single rule $R_\emptyset$, so V_A has one generator and dimension zero or one. The primitive bound specializes exactly to

$$r_A \leq B(1 + 0/\tau^2)^k = B.$$

□

Proposition A.2 (One pre-instance basis and exact coordinate map). *Under Assumptions 1, 2, and 4, and Proposition A.1, there is a basis $\psi_1, \dots, \psi_{r_A}$ of V_A , with the empty basis when $r_A = 0$, such that*

$$\varphi_A(x) := (\psi_1(x), \dots, \psi_{r_A}(x))$$

is fixed before every distribution, target, selected complete or valid response rule, and realized seed. Every $f \in V_A$ has a unique coordinate vector $a(f) \in \mathbb{R}^{r_A}$ satisfying

$$f(x) = \langle a(f), \varphi_A(x) \rangle \quad \text{for every } x \in \mathcal{X}.$$

The statement includes $r_A = 0$, empty $\mathcal{X}$ or $\mathcal{H}$, and $m = 0$.

Proof. Proposition A.1 makes V_A finite-dimensional with dimension r_A . Fix a linearly independent spanning list $\psi_1, \dots, \psi_{r_A}$; if $r_A = 0$, the list is empty and $V_A = \{0\}$.

For $r_A > 0$, spanning gives coefficients $a(f)_1, \dots, a(f)_{r_A}$ with

$$f = \sum_{j=1}^{r_A} a(f)_j \psi_j.$$

Linear independence makes the coefficients unique. Evaluation at $x \in \mathcal{X}$ gives the zero-residual identity

$$f(x) = \sum_{j=1}^{r_A} a(f)_j \psi_j(x) = \langle a(f), (\psi_1(x), \dots, \psi_{r_A}(x)) \rangle = \langle a(f), \varphi_A(x) \rangle.$$

The dependence order is

$$\begin{aligned} &(\mathcal{X}, A, \Omega_A, \mu_A, m, \tau, \text{complete public interface}) \\ &\longmapsto \mathcal{Q}_A \longmapsto \mathfrak{R}_A^{\text{all}} \longmapsto \{F_R\} \longmapsto V_A, r_A \\ &\longmapsto (\psi_1, \dots, \psi_{r_A}), \varphi_A. \end{aligned}$$

No $\mathcal{D}$, target, validity constraint, selected rule, or realized seed enters this chain. The seed law, rather than a realized seed, enters each F_R . Every later valid-rule family is a subset of the already fixed all-rule family.

If $r_A = 0$, then $f = 0$, $a(f)$, and $\varphi_A(x)$ are the empty vectors in $\mathbb{R}^0$, and their inner product is $0 = f(x)$. Empty $\mathcal{X}$ forces this branch and has no pointwise condition. Empty $\mathcal{H}$ leaves the construction unchanged and makes its later target-indexed use vacuous. At $m = 0$, the same basis is built from $\text{span}\{F_{R_\emptyset}\}$, with no query-round argument and with $r_A \leq B$.

Together with Lemma A.1 and Proposition A.1, this constructs the complete all-rule family, the exact average-before-span response space and rank, and one common coordinate map before every learning instance. $\square$

A.2 Compactness in One Fixed Function Space

Lemma A.2 (Evaluation coordinates determine the fixed topology). *Under Assumptions 2 and 4, and the conclusions of Propositions A.1 and A.2, suppose $r_A > 0$. Then*

$$\text{span}_{\mathbb{R}}\{\delta_x : x \in \mathcal{X}\} = V_A^*, \quad \delta_x(f) := f(x).$$

Consequently, there are points $z_1, \dots, z_{r_A} \in \mathcal{X}$ such that $\delta_{z_1}, \dots, \delta_{z_{r_A}}$ form a basis of V_A^ . The map*

$$E_Z : V_A \rightarrow \mathbb{R}^{r_A}, \quad E_Z(f) := (f(z_1), \dots, f(z_{r_A}))$$

is a linear isomorphism, and $\|f\|_Z := \|E_Z(f)\|_\infty$ is a norm inducing exactly the finite-dimensional topology in which K_A was defined.

Proof. Every δ_x is an algebraic linear functional on V_A . Put

$$W := \text{span}_{\mathbb{R}}\{\delta_x : x \in \mathcal{X}\} \subseteq V_A^*.$$

The basis $\psi_1, \dots, \psi_{r_A}$ from Proposition A.2 shows directly that $\dim V_A^* = r_A$: the functionals returning the coefficients in $f = \sum_j a(f)_j \psi_j$ form a basis of the dual.

Suppose that $W \neq V_A^*$, and write $s = \dim W < r_A$. Choose a basis $\lambda_1, \dots, \lambda_s$ of W . In the fixed basis of V_A , the equations

$$\lambda_i \left(\sum_{j=1}^{r_A} c_j \psi_j \right) = 0, \quad 1 \leq i \leq s,$$

form an s -by- r_A homogeneous real linear system. Row reduction has at most $s < r_A$ pivots, so the nullspace contains a nonzero vector c . Thus

$$f := \sum_{j=1}^{r_A} c_j \psi_j \neq 0, \quad \lambda_i(f) = 0 \quad (1 \leq i \leq s).$$

Every δ_x is a linear combination of the λ_i , so $f(x) = \delta_x(f) = 0$ for every $x \in \mathcal{X}$. Equality in $\mathbb{R}^{\mathcal{X}}$ is pointwise, so this makes $f = 0$, a contradiction. Hence $W = V_A^*$.

Starting with the empty list, whenever selected evaluations do not yet span V_A^* , choose another evaluation outside their span. Each addition raises the dimension by one, so after exactly r_A additions the selected evaluations form a basis. Denote their points by $z_1, \dots, z_{r_A}$. In particular, $r_A > 0$ forces $\mathcal{X} \neq \emptyset$.

The map E_Z is linear. If $E_Z(f) = 0$, every basis functional δ_{z_i} vanishes on f . Every δ_x is their linear combination, so $f(x) = 0$ for all x , hence $f = 0$. Thus E_Z is injective. Its domain and codomain both have dimension r_A , so it is surjective as well.

It follows that $\|f\|_Z = \|E_Z f\|_{\infty}$ is a norm. To identify its topology with the fixed topology, let $a(f)$ be the coordinate vector in the basis (ψ_j) , and set

$$M_{ij} := \psi_j(z_i), \quad 1 \leq i, j \leq r_A.$$

Then $E_Z(f) = Ma(f)$, and bijectivity of E_Z makes M invertible. Finite matrix multiplication gives

$$\|E_Z(f)\|_{\infty} \leq \left(\max_{1 \leq i \leq r_A} \sum_{j=1}^{r_A} |M_{ij}| \right) \|a(f)\|_{\infty}$$

and

$$\|a(f)\|_{\infty} \leq \left(\max_{1 \leq i \leq r_A} \sum_{j=1}^{r_A} |(M^{-1})_{ij}| \right) \|E_Z(f)\|_{\infty}.$$

Both multipliers are finite sums of fixed real entries. Hence the evaluation norm and the fixed-basis coordinate norm have the same open and closed sets. The latter is the finite-dimensional topology specified in the setup, so $\|\cdot\|_Z$ induces that same topology. These inequalities establish topological equivalence only; they assume no bound on the coefficients of F_R or of convex combinations. $\square$

Lemma A.3 (Heine–Borel in evaluation coordinates). *Under Assumptions 2 and 4, the conclusions of Propositions A.1 and A.2, and Lemma A.2, suppose $r_A > 0$. If $C \subseteq V_A$ is closed in the fixed finite-dimensional topology and bounded in $\|\cdot\|_Z$, then C is compact in that topology.*

Proof. Lemma A.2 makes E_Z an isometric linear homeomorphism from $(V_A, \|\cdot\|_Z)$ onto $(\mathbb{R}^{r_A}, \|\cdot\|_\infty)$. Hence $D := E_Z(C)$ is closed and bounded in $\mathbb{R}^{r_A}$. We prove compactness directly.

Every sequence in D has a convergent subsequence with limit in D . Boundedness places every coordinate sequence in an interval $[-Q, Q]$. A bounded real sequence has a convergent subsequence by bisection: bisect the interval, retain a closed half containing infinitely many terms, and iterate, choosing successively increasing indices from the retained halves. The nested intervals have lengths tending to zero, so completeness of $\mathbb{R}$ gives their unique common limit and convergence of the chosen subsequence. Apply this argument successively to coordinates $1, \dots, r_A$, each time taking a subsequence of the preceding one. The final subsequence converges in every coordinate and therefore in $\|\cdot\|_\infty$. Closedness of D keeps the limit in D .

This subsequence property first implies total boundedness. Otherwise, for some $\eta > 0$, one could recursively choose points whose pairwise distances are at least η ; that sequence has no convergent subsequence. Now let $\mathcal{U}$ be an open cover of D . If no $\delta > 0$ had the property that each relative ball $B_D(y, \delta)$ lies in some member of $\mathcal{U}$, one could choose $y_n \in D$ such that $B_D(y_n, 1/n)$ lies in no cover member. A convergent subsequence $y_{n_j} \rightarrow y$, together with one cover member containing y , contradicts this choice for all sufficiently large j . Such a δ therefore exists. A finite $\delta/2$ -net gives finitely many δ -balls, each contained in a member of the cover, and hence a finite subcover. Thus D is compact.

The continuous inverse E_Z^{-1} maps D onto C , proving that C is compact in the fixed topology of V_A . $\square$

Proposition A.3 (The fixed response body is compact and convex). *Under Assumptions 2 and 4, the conclusions of Lemma A.1 and Proposition A.1, and Lemmas A.2 and A.3, suppose $r_A > 0$. Then*

$$K_A = \overline{\text{conv}}\{F_R : R \in \mathfrak{R}_A^{\text{all}}\}$$

is a nonempty compact convex subset of V_A in the one fixed topology, and

$$K_A \subseteq \{f \in V_A : \|f\|_Z \leq 1\}.$$

Proof. Lemma A.1 makes the generating family nonempty. For every complete rule R and every selected point z_i , Proposition A.1 gives $|F_R(z_i)| \leq 1$. Hence

$$\|F_R\|_Z = \max_{1 \leq i \leq r_A} |F_R(z_i)| \leq 1.$$

If $f = \sum_{j=1}^N \alpha_j F_{R_j}$ is a finite convex combination, then for every i ,

$$|f(z_i)| \leq \sum_{j=1}^N \alpha_j |F_{R_j}(z_i)| \leq \sum_{j=1}^N \alpha_j = 1.$$

Thus the convex hull lies in the evaluation-norm unit ball. The reverse triangle inequality

$$|\|f\|_Z - \|g\|_Z| \leq \|f - g\|_Z$$

makes this ball closed. Since Lemma A.2 identifies its topology with the topology in the defining closure, K_A remains in that unit ball.

By definition, K_A is closed, and it is nonempty because it contains the nonempty convex hull. To verify convexity of the closure, let $f, g \in K_A$, $\theta \in [0, 1]$, and $\eta > 0$. Choose convex-hull elements f_0, g_0 satisfying $\|f - f_0\|_Z < \eta$ and $\|g - g_0\|_Z < \eta$. Then

$$\|\theta f + (1 - \theta)g - [\theta f_0 + (1 - \theta)g_0]\|_Z \leq \theta \|f - f_0\|_Z + (1 - \theta) \|g - g_0\|_Z < \eta.$$

Since $\theta f_0 + (1 - \theta)g_0$ belongs to the convex hull and η is arbitrary, $\theta f + (1 - \theta)g \in K_A$. Finally, Lemma A.3 turns the established closedness and boundedness into compactness in the same topology. $\square$

Lemma A.4 (Every point evaluation is continuous with exact reconstruction). *Under Assumptions 2 and 4, the conclusions of Propositions A.1 and A.2, and Lemma A.2, suppose $r_A > 0$. For every $x \in \mathcal{X}$, point evaluation $\delta_x(f) = f(x)$ is continuous in the fixed topology. If $e_1, \dots, e_{r_A}$ are the standard coordinate vectors and*

$$b_i(x) := (E_Z^{-1}e_i)(x),$$

then every $f \in V_A$ satisfies

$$f(x) = \sum_{i=1}^{r_A} b_i(x)f(z_i), \quad |f(x)| \leq \left(\sum_{i=1}^{r_A} |b_i(x)| \right) \|f\|_Z,$$

and

$$\delta_x(f) - (\delta_x \circ E_Z^{-1})(E_Z f) = 0.$$

Proof. Fix x . The composition $\ell_x := \delta_x \circ E_Z^{-1}$ is linear on $\mathbb{R}^{r_A}$. For $y = \sum_{i=1}^{r_A} y_i e_i$, linearity gives

$$\ell_x(y) = \sum_{i=1}^{r_A} y_i \ell_x(e_i) = \sum_{i=1}^{r_A} y_i b_i(x).$$

Therefore

$$|\ell_x(y)| \leq \sum_{i=1}^{r_A} |b_i(x)| |y_i| \leq \left(\sum_{i=1}^{r_A} |b_i(x)| \right) \|y\|_\infty.$$

The coefficient sum is finite for each fixed x , proving continuity. Substituting $y = E_Z f$ and using $E_Z^{-1} E_Z f = f$ yields both the reconstruction formula and the zero-residual factorization. No bound uniform in x is asserted or needed. $\square$

Proposition A.4 (The zero-dimensional and empty-domain branch). *Under Assumptions 2 and 4, and the conclusions of Lemma A.1 and Proposition A.1, suppose $r_A = 0$. Then*

$$V_A = K_A = \{0\}.$$

The empty evaluation map $E_\emptyset : V_A \rightarrow \mathbb{R}^0$ is a linear isomorphism and induces the unique zero-dimensional topology. The body K_A is nonempty, compact, and convex, and every point evaluation is the zero continuous functional. These conclusions include $\mathcal{X} = \emptyset$ and the zero-rank branch at $m = 0$.

Proof. A vector space of dimension zero contains only its zero vector, so $V_A = \{0\}$. Lemma A.1 makes the generator family nonempty, and every generator lies in V_A ; hence every generator is zero. Its convex hull and its closure are both $\{0\}$, proving $K_A = \{0\}$.

Both V_A and $\mathbb{R}^0$ are singleton vector spaces, so the empty evaluation map is directly linear, injective, and surjective. The singleton $\{0\}$ is convex and compact by the open-cover definition. Every point evaluation has domain $\{0\}$ and value zero, so it is continuous. If $\mathcal{X} = \emptyset$, there is no point evaluation to check. If $m = 0$, the unique empty response rule supplies the generator without any query or reply. $\square$

Consequences for the fixed response body. If $r_A > 0$, Lemma A.2 supplies actual domain evaluations and identifies their norm with the fixed topology; Proposition A.3 gives a nonempty compact convex K_A ; and Lemma A.4 makes every point evaluation continuous with exact reconstruction. If $r_A = 0$, Proposition A.4 proves all the same body and continuity conclusions directly. Thus the later minimax and intersection arguments always take place in one fixed compact body, including for empty $\mathcal{X}$ and $m = 0$. $\square$

A.3 Exact-Center Correlation and Nondegeneracy

Lemma A.5 (Exact-center legality). *Under Assumptions 1 and 2, and Lemma A.1, for every $\mathcal{D} \in \mathcal{P}(\mathcal{X})$ and $h \in \mathcal{H}$, define on every admitted public query-bearing history*

$$R_{\mathcal{D},h}^0(q_1, v_1, \dots, q_t) := \mathbb{E}_{x \sim \mathcal{D}} q_t(x, h(x)).$$

Then $R_{\mathcal{D},h}^0$ is a deterministic complete public-transcript rule in $\mathfrak{R}_A^{\text{all}}$ and belongs to $\mathfrak{R}_{A,\tau}(\mathcal{D}, h)$. Its validity deviation is zero at every round reached by every seed. For $m = 0$, it is the unique empty rule and validity is vacuous.

Proof. Fix $\mathcal{D}$ and h . At any admitted public query-bearing history, the current query is measurable, bounded, and $[-1, 1]$ -valued. Therefore its population expectation is well defined and obeys

$$-1 \leq \mathbb{E}_{x \sim \mathcal{D}} q_t(x, h(x)) \leq 1.$$

Since $\tau > 0$, $[-1, 1] \subseteq I_\tau$, so the displayed formula assigns an admissible reply to every admitted history, including histories not reached by one seed or preceding reply path.

For fixed $(\mathcal{D}, h)$, the formula is deterministic and takes as its runtime input only the public transcript, in particular the public query q_t ; it has no hidden-seed input. Lemma A.1 therefore identifies it as a complete rule in $\mathfrak{R}_A^{\text{all}}$.

On every reached seed path and at every reached round, its reply is the population center itself, so

$$|R_{\mathcal{D},h}^0(q_1, v_1, \dots, q_t) - \mathbb{E}_{x \sim \mathcal{D}} q_t(x, h(x))| = 0 \leq \tau.$$

This proves validity separately on every reached path. There is no accumulated tolerance error over the at most m rounds. For $m = 0$, there is no query-bearing history, and Lemma A.1 gives the unique empty rule. $\square$

Proposition A.5 (Exact seed-averaged correlation). *Under Assumptions 1, 2, and 3, and the conclusions of Proposition A.1 and Lemma A.5, for every $\mathcal{D} \in \mathcal{P}(\mathcal{X})$ and $h \in \mathcal{H}$,*

$$F_{R_{\mathcal{D},h}^0} \in \{F_R : R \in \mathfrak{R}_A^{\text{all}}\} \subseteq K_A \subseteq V_A$$

and

$$\begin{aligned} \mathbb{E}_{x \sim \mathcal{D}}[h(x) F_{R_{\mathcal{D},h}^0}(x)] &= 1 - 2 \mathbb{E}_{U \sim \mu_A} \mathcal{L}_{\mathcal{D},h}(g_{U, R_{\mathcal{D},h}^0}) \\ &\geq 1 - 2\varepsilon = \rho > \frac{1}{2}. \end{aligned}$$

The accuracy expectation is solely over the learner seed; no distribution over response policies and no favorable seed is introduced.

Proof. Lemma A.5 gives $R_{\mathcal{D},h}^0 \in \mathfrak{R}_A^{\text{all}}$. Hence Proposition A.1 defines its mean response and places it in the raw generating family. Every raw generator belongs to its convex hull and therefore to $K_A \subseteq V_A$; compactness is not needed for this inclusion.

The bounded Fubini theorem (Folland, 1999) applies with probability laws μ_A and $\mathcal{D}$, and with

$$a(u, x) := h(x)g_{u, R_{\mathcal{D}, h}^0}(x).$$

Assumption 2 and the setup supply the needed measurability, while $|a(u, x)| = 1$ gives integrability. Thus

$$\begin{aligned}\mathbb{E}_{x \sim \mathcal{D}}[h(x)F_{R_{\mathcal{D}, h}^0}(x)] &= \int_{\mathcal{X}} h(x) \left(\int_{\Omega_A} g_{u, R_{\mathcal{D}, h}^0}(x) \mu_A(du) \right) \mathcal{D}(dx) \\ &= \int_{\Omega_A} \left(\int_{\mathcal{X}} h(x)g_{u, R_{\mathcal{D}, h}^0}(x) \mathcal{D}(dx) \right) \mu_A(du).\end{aligned}$$

For every (u, x) , both binary labels lie in $\{-1, +1\}$, and the two possible cases give the pointwise identity

$$h(x)g_{u, R_{\mathcal{D}, h}^0}(x) = 1 - 2\mathbf{1}\{g_{u, R_{\mathcal{D}, h}^0}(x) \neq h(x)\}.$$

Integration over x gives

$$\int_{\mathcal{X}} h(x)g_{u, R_{\mathcal{D}, h}^0}(x) \mathcal{D}(dx) = 1 - 2\mathcal{L}_{\mathcal{D}, h}(g_{u, R_{\mathcal{D}, h}^0}),$$

and integration over the seed yields the exact identity in the statement.

Lemma A.5 proves that this particular deterministic complete rule is valid. The universal quantifier in Assumption 3 therefore applies to it and gives

$$\mathbb{E}_{U \sim \mu_A} \mathcal{L}_{\mathcal{D}, h}(g_{U, R_{\mathcal{D}, h}^0}) \leq \varepsilon.$$

Substitution, with no discarded term, gives

$$1 - 2\mathbb{E}_U \mathcal{L}_{\mathcal{D}, h}(g_{U, R_{\mathcal{D}, h}^0}) \geq 1 - 2\varepsilon = \rho > \frac{1}{2}.$$

The exact-center rule is only one rule already covered by the every-valid-rule premise; it does not replace the adversarial oracle model. Tolerance creates zero residual. At $m = 0$, the same calculation uses the empty rule and no query-round step. At $\varepsilon = 0$, nonnegativity of loss forces its seed expectation to be zero, so the correlation is exactly 1. $\square$

Proposition A.6 (Point-mass exclusion of zero rank). *Under Assumptions 1, 2, and 3, and the conclusions of Propositions A.1 and A.5, if $\mathcal{X} \neq \emptyset$ and $\mathcal{H} \neq \emptyset$, then for every $x_0 \in \mathcal{X}$ and $h \in \mathcal{H}$,*

$$h(x_0)F_{R_{\delta_{x_0}, h}^0}(x_0) \geq \rho > 0.$$

Consequently $V_A \neq \{0\}$ and $r_A \geq 1$.

Proof. For nonempty $\mathcal{X}$, the point mass δ_{x_0} belongs to $\mathcal{P}(\mathcal{X})$. Proposition A.5 with $\mathcal{D} = \delta_{x_0}$ gives

$$\begin{aligned}h(x_0)F_{R_{\delta_{x_0}, h}^0}(x_0) &= \mathbb{E}_{x \sim \delta_{x_0}}[h(x)F_{R_{\delta_{x_0}, h}^0}(x)] \\ &\geq \rho > \frac{1}{2} > 0.\end{aligned}$$

Thus this raw response is a nonzero function in V_A , and $r_A = \dim V_A \geq 1$. When $m = 0$, all these exact-center rules are the unique empty rule, but the point-mass conclusion is unchanged. If $\mathcal{X}$ or $\mathcal{H}$ is empty, no nonexistent point or target is selected, and the later sign condition is vacuous.

Together with Lemma A.5 and Proposition A.5, this proves that every distribution-target pair has an exact ρ -correlated member of the one fixed response body and that the zero-rank branch is impossible precisely in the nonempty-domain, nonempty-class regime. $\square$

A.4 Finite Simultaneous Margin by Minimax

Lemma A.6 (Finite-simplex minima occur at vertices). *Let $S \subseteq \mathcal{X}$ be nonempty and finite. For every real family $(a_x)_{x \in S}$,*

$$\min_{p \in \Delta(S)} \sum_{x \in S} p(x) a_x = \min_{x \in S} a_x,$$

where

$$\Delta(S) := \left\{ p \in \mathbb{R}^S : p(x) \geq 0 \text{ for all } x \in S, \sum_{x \in S} p(x) = 1 \right\}.$$

Both minima are attained at a simplex vertex, including when S is a singleton.

Proof. Choose $x_* \in S$ with $a_{x_*} = \min_{x \in S} a_x$. For every $p \in \Delta(S)$,

$$\sum_{x \in S} p(x) a_x \geq \sum_{x \in S} p(x) a_{x_*} = a_{x_*}.$$

At the vertex p^{x_*} , which assigns mass one to x_* , equality holds. This calculation also covers $S = \{x_*\}$, for which the simplex contains only that vertex. $\square$

Proposition A.7 (Sion's equality on the fixed response body). *Under Assumptions 2 and 4, and the conclusions of Propositions A.3 and A.4, Lemma A.4, and Lemma A.6, let $h \in \mathcal{H}$ and let $S \subseteq \mathcal{X}$ be nonempty and finite. On the fixed product $K_A \times \Delta(S)$, define*

$$L_h(f, p) := \sum_{x \in S} p(x) h(x) f(x).$$

Then

$$\max_{f \in K_A} \min_{p \in \Delta(S)} L_h(f, p) = \min_{p \in \Delta(S)} \max_{f \in K_A} L_h(f, p),$$

and every displayed maximum and minimum is attained.

Proof. We use Sion's Theorem 3.4 (Sion, 1958, Theorem 3.4) in its maximizing/minimizing orientation. The cited statement says that if M and N are nonempty convex compact spaces contained in linear topological spaces, and $G : M \times N \rightarrow \mathbb{R}$ is upper semicontinuous and quasiconcave in its first, maximizing variable for every fixed second variable, and lower semicontinuous and quasiconvex in its second, minimizing variable for every fixed first variable, then

$$\sup_{m \in M} \inf_{n \in N} G(m, n) = \inf_{n \in N} \sup_{m \in M} G(m, n). \quad (\text{A.1})$$

Proposition A.3 gives a nonempty compact convex K_A when $r_A > 0$, and Proposition A.4 gives the same conclusion directly when $r_A = 0$. Thus the maximizing set is always the same fixed body; it is not selected after h , S , or p .

Because S is nonempty and finite, $\Delta(S)$ contains every vertex, is convex, and is closed and bounded in the finite-dimensional linear topological space $\mathbb{R}^S$. The direct proof of Lemma A.3 applies with the exact substitutions $r_A \mapsto |S|$, $D \mapsto \Delta(S)$, and the selected evaluation coordinates replaced by the standard coordinates of $\mathbb{R}^S$. Coordinatewise bisection gives a convergent subsequence, closedness retains its limit, and the same finite-net and open-cover argument yields a finite subcover. Hence $\Delta(S)$ is compact. Thus both sets required by the cited theorem are nonempty, compact, and convex.

For fixed $p \in \Delta(S)$,

$$f \mapsto L_h(f, p) = \sum_{x \in S} p(x) h(x) \delta_x(f)$$

is a finite linear combination of point evaluations. Lemma A.4 gives their continuity in the positive-rank branch, and Proposition A.4 gives it in the zero-rank branch. Hence this map is continuous affine, therefore upper semicontinuous and quasiconcave, on K_A . For fixed $f \in K_A$,

$$p \mapsto L_h(f, p) = \sum_{x \in S} p(x) h(x) f(x)$$

is a finite linear function of the Euclidean coordinates of p . It is continuous affine, hence lower semicontinuous and quasiconvex, on $\Delta(S)$. The payoff is finite and real valued.

The mapping

$$M = K_A, \quad N = \Delta(S), \quad G = L_h$$

therefore discharges every hypothesis of Sion's theorem. Its conclusion, without reversing the players or changing signs, is

$$\sup_{f \in K_A} \inf_{p \in \Delta(S)} L_h(f, p) = \inf_{p \in \Delta(S)} \sup_{f \in K_A} L_h(f, p). \quad (\text{A.2})$$

The cited theorem supplies this order equality only; the fixed body, per-distribution correlation, vertex reduction, attainment, and later arbitrary-domain closure are established separately in this paper.

We now justify all four attainment assertions. For fixed p , continuity of $L_h(\cdot, p)$ on compact K_A gives a maximum. For fixed f , continuity of $L_h(f, \cdot)$ on compact $\Delta(S)$ gives a minimum; Lemma A.6 identifies an attaining vertex.

For the outer maximum, that lemma also gives

$$\inf_{p \in \Delta(S)} L_h(f, p) = \min_{x \in S} h(x) f(x).$$

This function is continuous in f , because for $f, g \in K_A$,

$$\left| \min_{x \in S} h(x) f(x) - \min_{x \in S} h(x) g(x) \right| \leq \max_{x \in S} |f(x) - g(x)|,$$

and the finitely many point evaluations are continuous. Compactness of K_A therefore makes the left outer supremum in (A.2) an attained maximum.

For the outer minimum, write $v(p) := \max_{f \in K_A} L_h(f, p)$. Every quantity $\max_{f \in K_A} |f(x)|$ is finite because evaluation is continuous on compact K_A . For $p, q \in \Delta(S)$,

$$\begin{aligned} |v(p) - v(q)| &\leq \max_{f \in K_A} |L_h(f, p) - L_h(f, q)| \\ &\leq \sum_{x \in S} |p(x) - q(x)| \max_{f \in K_A} |f(x)|. \end{aligned}$$

Thus v is continuous on compact $\Delta(S)$, so the right outer infimum in (A.2) is attained. Replacing all four suprema and infima by the attained maxima and minima proves the proposition. $\square$

Proposition A.8 (Exact-center lower bound for the fixed-body game). *Under Assumptions 1, 2, 3, and 4, and the conclusions of Propositions A.5 and A.7, let $h \in \mathcal{H}$ and let $S \subseteq \mathcal{X}$ be nonempty and finite. Then*

$$\min_{p \in \Delta(S)} \max_{f \in K_A} L_h(f, p) \geq \rho.$$

For each p , the exact-center rule and its mean response may depend on p , but every such response belongs to the one fixed K_A .

Proof. Fix $p \in \Delta(S)$. It defines the finitely supported law $\mathcal{D}_p \in \mathcal{P}(\mathcal{X})$ assigning mass $p(x)$ to $x \in S$ and zero elsewhere. Proposition A.5 with $(\mathcal{D}_p, h)$ gives $F_{R_{\mathcal{D}_p, h}}^0 \in K_A$ and

$$\begin{aligned} L_h(F_{R_{\mathcal{D}_p, h}}^0, p) &= \sum_{x \in S} p(x) h(x) F_{R_{\mathcal{D}_p, h}}^0(x) \\ &= \mathbb{E}_{x \sim \mathcal{D}_p} [h(x) F_{R_{\mathcal{D}_p, h}}^0(x)] \\ &\geq \rho. \end{aligned}$$

Consequently,

$$\max_{f \in K_A} L_h(f, p) \geq L_h(F_{R_{\mathcal{D}_p, h}}^0, p) \geq \rho. \quad (\text{A.3})$$

The quantifiers are

$$\forall p \in \Delta(S) \exists F_{R_{\mathcal{D}_p, h}}^0 \in K_A : L_h(F_{R_{\mathcal{D}_p, h}}^0, p) \geq \rho.$$

The witness may change with p , but the maximizing set does not: it is always the pre-instance body K_A . Taking the attained minimum of (A.3) over p preserves ρ and proves the claim. $\square$

Proposition A.9 (Attained simultaneous margin on every finite restriction). *Under Assumptions 1–4, and the conclusions of Lemma A.6 and Propositions A.7 and A.8, for every $h \in \mathcal{H}$ and every nonempty finite $S \subseteq \mathcal{X}$,*

$$\begin{aligned} \max_{f \in K_A} \min_{x \in S} h(x) f(x) &= \max_{f \in K_A} \min_{p \in \Delta(S)} L_h(f, p) \\ &= \min_{p \in \Delta(S)} \max_{f \in K_A} L_h(f, p) \\ &\geq \rho. \end{aligned}$$

There is an attaining $f_{h,S} \in K_A$ satisfying

$$h(x) f_{h,S}(x) \geq \rho \quad \text{for every } x \in S.$$

Proof. For fixed $f \in K_A$, apply Lemma A.6 to $a_x = h(x) f(x)$. This gives

$$\min_{p \in \Delta(S)} L_h(f, p) = \min_{x \in S} h(x) f(x).$$

Taking the attained maximum over the same fixed K_A proves the first equality. Proposition A.7 proves the second equality in the displayed orientation, and Proposition A.8 proves the final inequality at the unchanged threshold ρ .

The maximum is attained by Proposition A.7; call an attaining function $f_{h,S}$. Then

$$\min_{x \in S} h(x) f_{h,S}(x) \geq \rho,$$

which is equivalent, because $S \neq \emptyset$, to all the pointwise inequalities in the statement.

For $S = \{x_0\}$, the simplex contains the single point mass and the chain reduces to

$$\max_{f \in K_A} h(x_0) f(x_0) \geq h(x_0) F_{R_{\delta_{x_0}, h}}^0(x_0) \geq \rho.$$

The empty set is not passed to Sion's theorem; its constraint is handled as the empty intersection in Proposition A.10 below. At $\varepsilon = 0$, all equalities preserve the margin 1. $\square$

A.5 Arbitrary-Domain Exactification

Lemma A.7 (Closed signed constraints in the fixed body). *Under Assumptions 2 and 4, and the conclusions of Lemma A.4 and Proposition A.4, fix $h \in \mathcal{H}$ and $x \in \mathcal{X}$. Then*

$$C_{h,x} := \{f \in K_A : h(x)f(x) \geq \rho\}$$

is relatively closed in the one fixed body K_A , at the same threshold as Proposition A.9, including $\rho = 1$.

Proof. Let $\delta_x(f) = f(x)$. In the positive-rank branch, Lemma A.4 makes δ_x continuous in the fixed topology of V_A . In the zero-rank branch, Proposition A.4 makes it the zero continuous functional. Thus

$$T_{h,x} : K_A \rightarrow \mathbb{R}, \quad T_{h,x}(f) := h(x)\delta_x(f),$$

is continuous in every rank branch. The ray $[\rho, \infty)$ is closed in $\mathbb{R}$, and

$$C_{h,x} = T_{h,x}^{-1}([\rho, \infty)).$$

Therefore $C_{h,x}$ is closed in the subspace topology of the same K_A . The weak inequality and the numerical threshold are unchanged at $\rho = 1$. $\square$

Proposition A.10 (Finite feasibility in the fixed body). *Assume Assumptions 1, 2, 3, and 4. Under the nonemptiness conclusions of Propositions A.3 and A.4, together with Proposition A.9, fix $h \in \mathcal{H}$. Then for every finite $S \subseteq \mathcal{X}$, including $S = \emptyset$,*

$$\bigcap_{x \in S} C_{h,x} \neq \emptyset.$$

Thus $(C_{h,x})_{x \in \mathcal{X}}$ has the finite-intersection property inside one fixed K_A .

Proof. For $S = \emptyset$, the intersection over the empty subfamily equals its ambient set:

$$\bigcap_{x \in \emptyset} C_{h,x} = K_A.$$

Proposition A.3 in the positive-rank branch and Proposition A.4 in the zero-rank branch make this set nonempty.

Now suppose $S \neq \emptyset$. Proposition A.9 supplies $f_{h,S} \in K_A$ such that

$$h(x)f_{h,S}(x) \geq \rho \quad \text{for every } x \in S.$$

By definition, $f_{h,S} \in C_{h,x}$ for every $x \in S$, and therefore

$$f_{h,S} \in \bigcap_{x \in S} C_{h,x}.$$

These two cases exhaust all finite subfamilies. The witnesses for different finite sets need not agree, be nested, or converge. $\square$

Lemma A.8 (Compactness closes an arbitrary finite-intersection family). *Let K be a nonempty compact topological space, let I be an arbitrary, possibly empty, index set, and let $(D_i)_{i \in I}$ be relatively closed subsets of K . If*

$$\bigcap_{i \in J} D_i \neq \emptyset \quad \text{for every finite } J \subseteq I,$$

where the intersection for $J = \emptyset$ is K , then

$$\bigcap_{i \in I} D_i \neq \emptyset.$$

Proof. Suppose instead that $\bigcap_{i \in I} D_i = \emptyset$. Taking complements relative to K gives

$$K = K \setminus \bigcap_{i \in I} D_i = \bigcup_{i \in I} (K \setminus D_i).$$

Each $K \setminus D_i$ is relatively open, so these complements form an open cover of K . Compactness supplies finitely many indices $i_1, \dots, i_n$ whose complements already cover K . Since $K \neq \emptyset$, an empty subcollection cannot cover it, so $n \geq 1$. Taking complements again gives

$$\bigcap_{j=1}^n D_{i_j} = \emptyset,$$

contradicting finite-intersection nonemptiness. Hence the total intersection is nonempty. The argument uses the open-cover definition of compactness and does not enumerate I , pass to a sequence, or impose a cardinality condition. If $I = \emptyset$, the conclusion is simply $K \neq \emptyset$. $\square$

Proposition A.11 (One exact-margin witness on the arbitrary domain). *Under Assumptions 1–4, and the conclusions of Propositions A.3, A.4, A.9, and A.10, and Lemmas A.4, A.7, and A.8, for every $h \in \mathcal{H}$ there exists $f_h \in K_A$ such that*

$$h(x)f_h(x) \geq \rho \quad \text{for every } x \in \mathcal{X}.$$

The conclusion holds for an arbitrary, possibly uncountable or empty, domain and preserves ρ exactly, including $\rho = 1$.

Proof. Fix $h \in \mathcal{H}$. Proposition A.3 in the positive-rank branch and Proposition A.4 in the zero-rank branch make the same K_A nonempty and compact. Lemma A.7 makes every $C_{h,x}$ relatively closed in that body, and Proposition A.10 gives the finite-intersection property, including the empty subfamily. Apply Lemma A.8 with

$$K = K_A, \quad I = \mathcal{X}, \quad D_x = C_{h,x}.$$

It follows that

$$\bigcap_{x \in \mathcal{X}} C_{h,x} \neq \emptyset.$$

Choose f_h in this intersection. For every x , its membership in $C_{h,x}$ is exactly $h(x)f_h(x) \geq \rho$.

No relationship among the finite witnesses $f_{h,S}$ was used. They may vary arbitrarily with S ; their only role is to certify finite intersections inside the same K_A , while compactness produces f_h .

If $\mathcal{X} = \emptyset$, the indexed family is empty and its full intersection is K_A ; the pointwise condition is vacuous. If $\mathcal{H} = \emptyset$, the outer target quantifier is vacuous. In the zero-rank branch, $K_A = \{0\}$. Empty $\mathcal{X}$ is compatible with this branch. If instead $\mathcal{X} \neq \emptyset$ and some $h \in \mathcal{H}$ exists, Proposition A.9 applied to $S = \{x\}$ would require $h(x)0 \geq \rho > 1/2$, a contradiction. Thus the preceding results themselves exclude zero rank exactly in the nonempty target/domain regime; no positive-rank assumption was added.

At $\rho = 1$, every finite witness belongs to the closed sets defined by $h(x)f(x) \geq 1$, and the compact intersection uses those identical sets. No relaxation, sequential approximation, or smaller positive threshold is introduced. This proves the arbitrary-domain conclusion at zero residual. $\square$

A.6 Exact Scores and Polynomial Dimension Closure

Proposition A.12 (Exact score transfer through the pre-instance basis). *Under Assumptions 1–4, and the conclusions of Propositions A.2 and A.11, the basis $\psi_1, \dots, \psi_{r_A}$ and map φ_A are fixed before every learning instance. For every $h \in \mathcal{H}$ and every $f_h \in K_A$ supplied by Proposition A.11, there is a unique $w_h \in \mathbb{R}^{r_A}$ such that*

$$f_h = \sum_{j=1}^{r_A} (w_h)_j \psi_j$$

and, for every $x \in \mathcal{X}$,

$$\langle w_h, \varphi_A(x) \rangle = f_h(x), \quad h(x) \langle w_h, \varphi_A(x) \rangle \geq \rho > \frac{1}{2} > 0.$$

Thus every nonvacuous score is nonzero and has sign $h(x)$. Only f_h and w_h may depend on h .

Proof. Proposition A.2 constructs the basis and φ_A from the all-rule mean-response space, before any distribution, target, validity restriction, selected rule, or realized seed. Fixing a later target cannot change this basis or map.

Now fix $h \in \mathcal{H}$. Proposition A.11 supplies $f_h \in K_A$ with

$$h(x) f_h(x) \geq \rho \quad \text{for every } x \in \mathcal{X}.$$

The closure defining K_A is taken inside V_A , so $f_h \in V_A$. The basis property gives a unique $w_h \in \mathbb{R}^{r_A}$ satisfying

$$f_h = \sum_{j=1}^{r_A} (w_h)_j \psi_j.$$

Evaluating at any x gives

$$\begin{aligned} f_h(x) &= \sum_{j=1}^{r_A} (w_h)_j \psi_j(x) \\ &= \langle w_h, (\psi_1(x), \dots, \psi_{r_A}(x)) \rangle \\ &= \langle w_h, \varphi_A(x) \rangle. \end{aligned}$$

This is an equality, with no limit, norm conversion, approximation, or coordinate residual. Multiplying by $h(x)$ transfers the global margin:

$$h(x) \langle w_h, \varphi_A(x) \rangle = h(x) f_h(x) \geq \rho.$$

Assumption 1 gives $\rho > 1/2 > 0$. Hence when $h(x) = +1$ the score is positive, and when $h(x) = -1$ it is negative.

If $\mathcal{H} = \emptyset$, the target statement is vacuous and the map remains fixed. If $\mathcal{X} = \emptyset$, all pointwise assertions are vacuous; Proposition A.1 gives $V_A = \{0\}$ and $r_A = 0$, so $f_h = 0$, w_h is the empty vector, and the basis expansion is the empty sum. $\square$

Lemma A.9 (Dimension-complexity admissibility and degenerate branches). *Assume Assumptions 1, 2, 3, and 4. Under the conclusions of Proposition A.12 and Proposition A.6,*

$$\text{dc}(\mathcal{H}) \leq r_A.$$

If $\mathcal{X} = \emptyset$ or $\mathcal{H} = \emptyset$, then $\text{dc}(\mathcal{H}) = 0$. If $r_A = 0$, at least one of these two sets is empty and $\text{dc}(\mathcal{H}) = r_A = 0$.

Proof. Proposition A.12 gives one fixed map

$$\varphi_A : \mathcal{X} \rightarrow \mathbb{R}^{r_A}$$

such that for every $h \in \mathcal{H}$, some $w_h \in \mathbb{R}^{r_A}$ satisfies

$$h(x)\langle w_h, \varphi_A(x) \rangle > 0 \quad \text{for every } x \in \mathcal{X}.$$

Thus $d = r_A$ belongs to the set of admissible dimensions in the definition of $\text{dc}(\mathcal{H})$, and taking its infimum gives $\text{dc}(\mathcal{H}) \leq r_A$. The admissible set is nonempty, so its empty-infimum convention is not invoked under these assumptions.

If $\mathcal{H} = \emptyset$, dimension zero is admissible: use the unique map from $\mathcal{X}$ to $\mathbb{R}^0$, and the target-indexed requirement is vacuous. Since admissible dimensions lie in $\mathbb{N}_0$, this gives $\text{dc}(\mathcal{H}) = 0$.

If $\mathcal{X} = \emptyset$, dimension zero is again admissible. Use the unique map $\emptyset \rightarrow \mathbb{R}^0$; for every target, choose the empty weight, and the pointwise condition is vacuous. Hence $\text{dc}(\mathcal{H}) = 0$. In this branch, Proposition A.1 additionally gives $r_A = 0$.

Finally, suppose $r_A = 0$. If both $\mathcal{X}$ and $\mathcal{H}$ were nonempty, Proposition A.6 would give $r_A \geq 1$, a contradiction. Therefore $\mathcal{X} = \emptyset$ or $\mathcal{H} = \emptyset$, and the preceding paragraphs give $\text{dc}(\mathcal{H}) = 0 = r_A$. This exhausts the zero-rank boundary without asking a zero-dimensional score to realize a nonempty sign constraint. $\square$

Proposition A.13 (Exact conditional polynomial dimension closure). *Under Assumptions 1–4, and the conclusions of Proposition A.12 and Lemma A.9, the fixed map $\varphi_A : \mathcal{X} \rightarrow \mathbb{R}^{r_A}$ satisfies, for every $h \in \mathcal{H}$, the existence of $w_h \in \mathbb{R}^{r_A}$ with*

$$h(x)\langle w_h, \varphi_A(x) \rangle \geq 1 - 2\varepsilon = \rho > \frac{1}{2} > 0 \quad \text{for every } x \in \mathcal{X},$$

and

$$\text{dc}(\mathcal{H}) \leq r_A \leq B(1 + m/\tau^2)^k.$$

The conclusion is deterministic and contains no hidden constants. It remains conditional on the primitive mean-response-rank certificate and makes no claim about seed-specific terminal rank, a rank-free implication, an unconditional theorem, or a universal linear bound.

Proof. The pointwise score statement, including pre-instance map independence and strict positivity, is Proposition A.12. Lemma A.9 gives

$$\text{dc}(\mathcal{H}) \leq r_A.$$

Assumption 4 gives, for the same r_A ,

$$r_A \leq B(1 + m/\tau^2)^k.$$

Transitivity proves the displayed chain. Both inequalities are retained verbatim; no term is absorbed, asymptotic notation introduced, or constant hidden.

The parameter boundaries follow by direct substitution. If $m = 0$, then

$$B(1 + m/\tau^2)^k = B(1 + 0)^k = B,$$

so $\text{dc}(\mathcal{H}) \leq r_A \leq B$. The unique empty complete response rule from Lemma A.1 applies, and no query-round argument is used. If $\varepsilon = 0$, then $\rho = 1$ and

$$h(x)\langle w_h, \varphi_A(x) \rangle \geq 1$$

with no margin degradation or probabilistic surrogate. If $B = 1$, the rank bound is

$$r_A \leq (1 + m/\tau^2)^k;$$

if $k = 1$, it is

$$r_A \leq B(1 + m/\tau^2);$$

and if $B = k = 1$, it is exactly

$$r_A \leq 1 + m/\tau^2.$$

These substitutions use only $m \in \mathbb{N}_0$, $\tau > 0$, $B \geq 1$, and $k \geq 1$, and introduce no upper bound on τ .

Empty $\mathcal{X}$, empty $\mathcal{H}$, and $r_A = 0$ are handled by Lemma A.9. The primitive certificate, rather than the number of response transcripts, terminal predictors, or queries alone, supplies the second dimension inequality. The probability mode is deterministic, the horizon mode is the fixed finite upper horizon m , and the metric consists of exact algebraic dimension and the pointwise signed score. Thus every exposed dependence is the one displayed above. $\square$

A.7 Proof of the Main Theorem

Proof of Theorem 3.1. Lemma A.1 and Propositions A.1 and A.2 construct the fixed all-rule mean-response space, its finite rank, and one pre-instance coordinate map. Propositions A.3 and A.4, together with Lemma A.4, place all subsequent arguments in one nonempty compact convex body with continuous point evaluations.

Lemma A.5 and Proposition A.5 produce the exact distributionwise signed-correlation witness, while Proposition A.6 settles the nonempty zero-rank boundary. Proposition A.9 converts those witnesses, through the fixed-body minimax equality, into an attained simultaneous margin on every nonempty finite restriction. Proposition A.11 then uses the closed constraints and compact finite-intersection argument to obtain one f_h on the entire arbitrary domain.

Finally, Proposition A.12 converts f_h to exact basis scores with zero residual, Lemma A.9 proves the dimension and degenerate-case conclusions, and Proposition A.13 appends the primitive rank certificate and the exact parameter specializations. These statements are precisely all conclusions of Theorem 3.1. $\square$